ECA18 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO4 – INVESTIGATIVE ASSIGNMENT REPORT

Ultra-High-Speed Photonic Transport Link: Technology Evaluation, Impairment Mitigation, and Integrated System Design

Student Information	Details
Student Name	P.ROHITH
Register Number	192312015
Course Code	ECA18
Course Title	Optoelectronics and Optical Communication
Course Outcome	CO4
Assessment Type	Investigative Assignment
Duration	1 Week
Maximum Marks	100

1. COURSE OUTCOME

Design and analyze modulation formats (OOK, PAM4, DPSK, QAM, coherent QAM), implement modulators (EO, AO, thermo-optic, MZI, ring resonators), and evaluate optical amplifiers (EDFA, SOA, Raman) in WDM systems with DSP compensation.

2. ENGINEERING INVESTIGATION PROBLEM

A next-generation optical network operator plans to deploy a high-capacity Dense Wavelength Division Multiplexing (DWDM) optical transport link. The system must achieve high spectral efficiency, robust transmission quality, low noise, high integration scalability and standards-oriented implementation. The investigation evaluates advanced modulation formats, high-speed photonic modulators, polarization management, optical amplification and coherent DSP-based impairment compensation.

The design target is a 32-channel DWDM system with 200 Gb/s per channel. Candidate modulation formats include OOK, PAM4, DPSK, DQPSK, QAM and coherent 16-QAM/64-QAM. Candidate photonic platforms include MZI and micro-ring structures using silicon photonics and EO polymers. Candidate amplification includes EDFA, SOA, Raman and hybrid Raman–EDFA. Major impairments are CD, PMD, Kerr nonlinearities, WDM crosstalk and ASE noise. The source assignment requires OptiSystem/OptSim/MATLAB analytical and simulation validation.

3. GIVEN PARAMETERS AND CONSTRAINTS

Constraint / Parameter	Specification
WDM configuration	32-channel DWDM; ITU-T G.698 grid spacing
Target channel rate	200 Gb/s per DWDM channel
Candidate modulation	OOK (NRZ/RZ), PAM4, DPSK, DQPSK, QAM, coherent 16-QAM / 64-QAM
Modulator mechanisms	EO, AO, thermo-optic, magneto-optic, Kerr effect
Modulator architectures	MZI, micro-ring resonator, EO polymers, silicon photonics
Polarization management	Isolators, circulators, Faraday rotators, PMD compensators
Amplifiers	EDFA, SOA, distributed/lumped Raman, hybrid EDFA + Raman
Coupling / multiplexing	Directional couplers, MMI, AWG
Major impairments	CD, PMD, SPM, XPM, FWM, WDM crosstalk, ASE
DSP	Linear equalization, polarization demultiplexing, carrier phase estimation, DBP
Validation	ITU-T G.698, IEEE 802.3ct/ck, OIF-CEI; OptiSystem / OptSim / MATLAB

4. INFORMATION NOT GIVEN – ENGINEERING DECISIONS REQUIRED

- Final modulation format and symbol rate.

- Specific modulator mechanism, waveguide architecture and material platform.
- Optical amplifier configuration, gain and gain-flatness profile.
- Primary optical versus digital impairment-compensation strategy.
- Sequence and algorithm structure of coherent DSP.
- Final integrated WDM architecture and deployment roadmap.

The source assignment explicitly requires these decisions to be established through analytical calculations, technology comparison, simulation and engineering judgment. Where numerical span power values are not supplied in the source, the calculations below use clearly stated engineering assumptions rather than presenting them as given values.

STAGE 1 – INTERPRET THE COMMUNICATION REQUIREMENT

The first design step is to translate the 32-channel, 200 Gb/s/channel requirement into aggregate capacity and physical-layer targets.

Aggregate capacity

Ctotal = N × Rch = 32 × 200 Gb/s

Therefore, Ctotal = 6.4 Tb/s

- The system must support 6.4 Tb/s gross aggregate channel capacity.
- Spectral efficiency should be maximized to keep the occupied optical spectrum and channel spacing manageable.
- OSNR, pre-FEC BER and Q-factor must remain within the selected receiver/FEC operating margin.
- Launch power must be optimized because increasing power can improve OSNR initially but eventually increases Kerr nonlinearities.
- Modulator bandwidth must accommodate the selected symbol rate plus practical roll-off.
- Insertion losses from modulators, couplers and AWGs must be included in the power budget.
- CD and PMD require receiver DSP compensation; nonlinear effects require optimized launch power and, where justified, DBP.
- Photonic integration must balance silicon-photonics compactness with thermal sensitivity and modulator efficiency.
- Reliability, power consumption and thermal management become increasingly important as channel count increases.

Investigation Requirement Statement: Design a 32-channel, 6.4 Tb/s DWDM link using a spectrally efficient coherent modulation format, integrated high-speed photonic modulators, flattened optical amplification, polarization management and coherent DSP capable of compensating the dominant linear and nonlinear impairments.

STAGE 2 – INVESTIGATE AND SELECT THE MODULATION / MODULATOR CONFIGURATION

Format	Bits/symbol (per polarization)	200 Gb/s symbol rate	Main advantage	Main limitation
NRZ-OOK	1	200 GBd	Simple transmitter/receiver	Poor spectral efficiency at this capacity
PAM4	2	100 GBd	Lower baud rate than OOK	High electrical bandwidth; tighter SNR margin
DPSK	1	200 GBd	Phase-based detection benefits	Lower spectral efficiency
DQPSK	2	100 GBd	Lower baud rate	Complexity and nonlinear/OSNR trade-offs
DP-16QAM	8 total with dual polarization	25 GBd	High spectral efficiency; coherent DSP	Higher OSNR and DSP complexity
DP-64QAM	12 total with dual polarization	16.67 GBd	Very high spectral efficiency	Much tighter OSNR/implementation margin

For dual-polarization 16-QAM, each symbol carries 4 bits on each polarization, giving 8 bits/symbol across both polarizations.

DP-16QAM baud rate

$R_s = R_b / (2 \times \log_2(16)) = 200\ \text{Gb/s} / (2 \times 4)$

Therefore, R_s = 25 Gbaud

PAM4 baud rate

$R_s = R_b / \log_2(4) = 200\ \text{Gb/s} / 2$

Therefore, R_s = 100 Gbaud

DP-64QAM baud rate

$R_s = 200\ \text{Gb/s} / (2 \times \log_2(64)) = 200 / 12$

Therefore, $R_s \approx$ 16.67 Gbaud

With a practical raised-cosine roll-off factor $\alpha = 0.20$, an approximate occupied bandwidth is $B \approx (1+\alpha)R_s$. Thus DP-16QAM requires about 30 GHz and can fit within a 50 GHz-class channel grid with engineering margin, while PAM4 requires about 120 GHz and therefore consumes much more spectrum.

Modulation mechanisms: electro-optic modulation is the preferred high-speed mechanism. Acousto-optic and thermo-optic mechanisms are generally too slow for direct 25–100 GBd data modulation, while magneto-optic and Kerr mechanisms are more specialized. EO modulation using MZI or compact resonant structures is therefore the practical choice.

Architecture	Bandwidth / Speed	Footprint	Thermal sensitivity	Insertion loss / Design issue
Silicon MZI	High; suitable for high-speed coherent data	Moderate	Moderate; heater tuning may be required	Good bandwidth, relatively larger than rings
Silicon MRR	Very compact; resonant bandwidth dependent	Very small	High; resonance shifts with temperature	Low footprint but tighter thermal control
EO-polymer MZI	Potentially very high speed and low $V\pi$	Moderate	Material stability must be considered	Potentially low drive voltage
EO-polymer ring	Compact and fast potential	Very small	Resonance/material stability important	Strong fabrication and thermal sensitivity

DESIGN DECISION 1 – SELECT THE PROPOSED MODULATION / MODULATOR CONFIGURATION

I recommend DP-16QAM with a high-speed electro-optic MZI modulator implemented on a silicon-photonic platform because it carries 200 Gb/s at only 25 Gbaud, has substantially lower baud rate and spectral occupancy than PAM4, and provides a practical compromise between spectral efficiency, OSNR requirement, DSP complexity and photonic integration. A thermally stabilized resonant MRR may be considered where footprint is the dominant constraint, but MZI provides a more forgiving wideband operating architecture.

STAGE 3 – DEVELOP AND EVALUATE THE OPTICAL AMPLIFICATION CONFIGURATION

Amplifier	Gain bandwidth	Output power	Noise / gain flatness	Suitability
EDFA	Wide C-band	High	Good, but ASE and gain tilt accumulate	Primary long-haul amplifier
SOA	Broad; compact	Moderate	Higher noise and nonlinear effects	PIC/short-reach amplification
Distributed Raman	Broad and wavelength-flexible	Span-dependent	Excellent noise-performance potential	Improves OSNR by distributed gain
Lumped Raman	Broad	Moderate/high	Can improve span budget	Useful supplementary stage
Hybrid Raman + EDFA	Wide C-band	High	Low effective noise with flattened gain	Recommended for high-capacity DWDM

The source assignment does not provide numerical per-channel input and target output powers in the extracted problem statement. Therefore, the following span calculation is an explicit engineering assumption for demonstrating the required method: per-channel amplifier input = −10 dBm and target per-channel output = 0 dBm.

Per-channel span gain

$G_{ch} = P_{out,ch} - P_{in,ch} = 0 - (-10)$

Therefore, $G_{ch} = 10$ dB

Total amplifier output power

$P_{total} = 32 \times 1\ mW = 32\ mW$

Therefore, $P_{total} = 0.032\ W \approx 15.05$ dBm

This 15.05 dBm total is the summed optical output for 32 channels at 0 dBm/channel. The actual deployed gain must be recalculated from the measured span loss, connector loss, AWG/MMI loss, fiber attenuation, design margin and required receiver power.

EDFA gain is not perfectly flat across the C-band. A gain-flattening filter (GFF), dynamic gain equalization or per-channel power control should therefore be included. Raman pre-amplification can improve effective noise performance, while EDFA supplies practical high-output power and mature C-band operation.

ASE accumulates with cascaded amplifiers and progressively reduces OSNR. The design should therefore minimize unnecessary amplifier stages, use low-noise Raman assistance where reach requires it, and equalize channels before nonlinear effects become excessive.

DESIGN DECISION 2 – SELECT THE PROPOSED AMPLIFICATION CONFIGURATION

I recommend distributed Raman assistance followed by a gain-flattened EDFA because the hybrid architecture can improve span OSNR while retaining the high output power and mature C-band operation of EDFA. GFF/dynamic equalization should control spectral tilt, and launch power should be optimized to avoid converting the OSNR improvement into excessive Kerr nonlinear penalty.

STAGE 4 – INVESTIGATE AND SELECT THE WDM IMPAIRMENT / DSP COMPENSATION CONFIGURATION

Impairment	Physical mechanism	Effect	Preferred mitigation
CD	Different wavelengths propagate at different group velocities	Pulse/symbol spreading and ISI	Static CD equalization using FIR/FDE
PMD	Polarization-dependent group delay	Polarization distortion and time-varying ISI	Adaptive MIMO equalizer
SPM	Signal intensity changes its own phase through Kerr effect	Spectral broadening and phase distortion	Launch-power optimization + DBP
XPM	One WDM channel modulates phase of another	Inter-channel phase distortion	Power optimization + DBP
FWM	Three optical frequencies generate new mixing products	Crosstalk and ghost channels	Channel spacing/power optimization + DSP where possible
PDL	Polarization-dependent insertion loss	Polarization imbalance	Polarization tracking/equalization
ASE	Amplifier spontaneous emission	OSNR degradation	Low-NF amplification and gain management
WDM crosstalk	Finite isolation in mux/demux devices	Inter-channel interference	AWG/MMI design and filtering

Optical DCF can compensate CD but introduces extra loss and nonlinear/dispersion-management penalties. Optical phase conjugation can compensate some propagation distortions but increases system complexity. Coherent digital compensation is preferred because it is programmable, adaptable and avoids inserting a large amount of additional passive fiber.

Recommended coherent DSP sequence: ADC → resampling/timing recovery → chromatic-dispersion compensation → adaptive 2×2 polarization MIMO equalizer → carrier-frequency recovery → carrier-phase estimation → nonlinear compensation/DBP where required → symbol decision → FEC.

DBP uses the split-step Fourier method to numerically invert the nonlinear Schrödinger propagation model. More steps and more spatial resolution can improve compensation but increase computational complexity and power consumption. Therefore, reduced-step DBP or perturbation-based nonlinear compensation is preferable when full DBP is too expensive.

DESIGN DECISION 3 – SELECT THE DSP COMPENSATION CONFIGURATION

I recommend coherent DSP with frequency-domain CD compensation, adaptive 2×2 MIMO polarization equalization, carrier recovery and reduced-complexity digital backpropagation because this combination directly addresses CD, PMD and major Kerr nonlinear impairments while avoiding the additional loss and complexity of extensive optical dispersion-compensating fiber.

STAGE 5 – EVALUATE THE INTEGRATED WDM TRANSMISSION STRATEGY

Total throughput
32 channels × 200 Gb/s
Therefore, 6.4 Tb/s

Approximate DP-16QAM spectral efficiency
200 Gb/s / 50 GHz
Therefore, 4 bit/s/Hz per channel on a 50 GHz grid

The 4 bit/s/Hz value is the gross channel-rate-to-grid-spacing figure; practical net spectral efficiency is lower after FEC overhead, guard bands and implementation penalties.

- Higher-order modulation reduces baud rate and spectral occupancy but increases OSNR sensitivity.
- The lower 25 GBd DP-16QAM symbol rate eases modulator bandwidth requirements compared with 100 GBd PAM4.
- Higher launch power improves received OSNR only until nonlinear effects such as SPM and XPM become dominant.
- MMI/AWG insertion loss reduces available launch/receiver power and must be included in the amplifier power budget.
- Polarization management and coherent MIMO DSP are essential for a dual-polarization architecture.
- A hybrid Raman–EDFA chain provides an attractive OSNR/power compromise for long spans.
- Silicon photonics supports high integration density, but thermal stabilization and packaging are important.
- Scalability depends on channel control, laser stability, modulator yield, DSP processing power and amplifier gain management.

DESIGN DECISION 4 – CREATE THE INTEGRATED WDM STRATEGY

I recommend adopting the integrated DP-16QAM DWDM strategy with silicon-photonic EO MZI transmitters, AWG/MMI multiplexing, polarization management, hybrid Raman–EDFA spans, AWG demultiplexing and a coherent ICR/DSP receiver because it provides a balanced 6.4 Tb/s architecture with practical baud rate, high spectral efficiency and programmable impairment compensation.

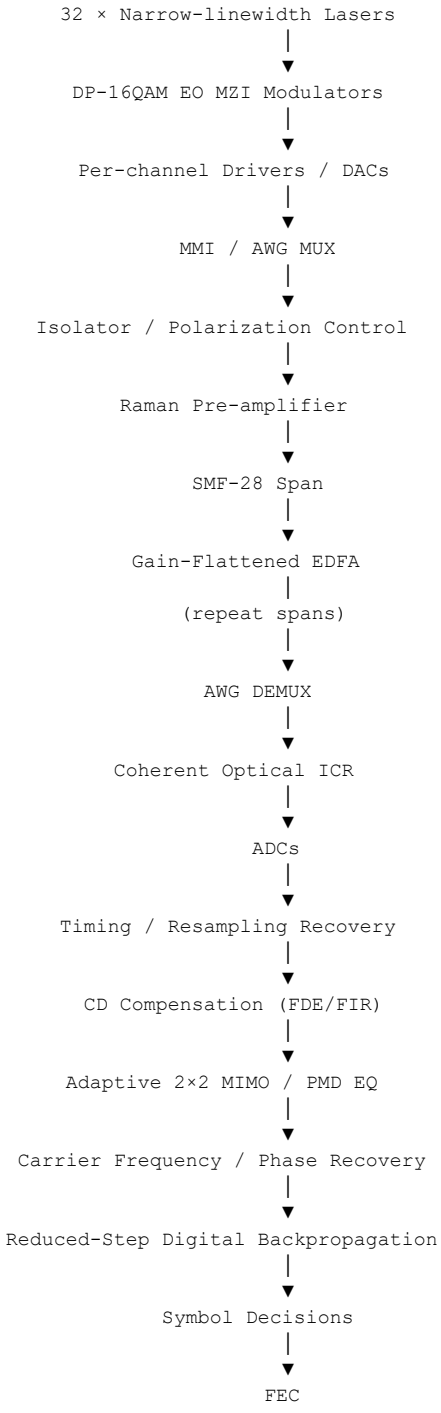
STAGE 6 – VERIFY AND CRITICALLY EVALUATE THE COMPLETE WDM SYSTEM

Requirement / Performance Parameter	Target Value	Achieved / Evidence
Aggregate WDM Capacity	6.4 Tb/s	6.4 Tb/s = 32 × 200 Gb/s
Spectral efficiency / modulation	DP-16QAM	25 Gbaud; ~4 bit/s/Hz gross on 50 GHz grid
Modulator bandwidth / footprint	High-speed, compact PIC	25 GBd channel; EO MZI selected
Amplifier gain / flatness	Span-loss dependent; flat C-band	10 dB illustrative gain under stated assumption; GFF recommended
Total amplifier output	32 channels	32 mW / 15.05 dBm for 0 dBm/ch assumption
OSNR / BER	Pre-FEC margin required	Improved using Raman + EDFA, launch optimization and coherent DSP
CD / PMD mitigation	Full link tolerance	FDE + adaptive MIMO
Nonlinear mitigation	SPM/XPM/FWM tolerance	Launch optimization + reduced-step DBP
Standards / feasibility	ITU-T G.698; IEEE 802.3ct/ck; OIF-CEI	Architecture aligned to source-stated standards; compliance requires detailed implementation verification

The achieved values above are analytical design values. The source assignment requires OptiSystem/OptSim/MATLAB simulation evidence such as eye/constellation plots, optical spectra and Q-factor versus launch power. Those plots cannot be honestly claimed as measured simulation results unless the model is actually run in the specified simulator.

FINAL SYSTEM DESIGN – K6 CREATE / DESIGN

The proposed end-to-end architecture is:



FINAL SYSTEM DESIGN PARAMETER TABLE

Parameter	Adopted Design
Number of WDM channels	32
Per-channel gross rate	200 Gb/s
Aggregate gross rate	6.4 Tb/s
Modulation	Dual-polarization 16-QAM
Symbol rate	25 Gbaud
Example channel grid	50 GHz class

Gross grid spectral efficiency	4 bit/s/Hz
Modulator	High-speed EO MZI
PIC platform	Silicon photonics
Polarization management	Isolators + polarization control; adaptive MIMO at receiver
Multiplexer / demultiplexer	AWG / MMI-based
Amplification	Distributed Raman + gain-flattened EDFA
Illustrative span gain	10 dB, based on assumed −10 to 0 dBm/channel
Illustrative total output	32 mW = 15.05 dBm
Fiber	SMF-28 class single-mode fiber
CD compensation	Frequency-domain/FIR equalizer
PMD compensation	Adaptive 2×2 MIMO
Nonlinear compensation	Reduced-step DBP
Receiver	Coherent integrated receiver + ADC
Validation	OptiSystem / OptSim / MATLAB analytical model

MATLAB ANALYTICAL / SIMULATION MODEL

The following MATLAB script provides a starting model for modulation-rate comparison, WDM spectrum placement, amplifier power summation and a simplified Q-factor versus launch-power trend. It is an analytical model, not a substitute for a full OptiSystem/OptSim physical simulation.

```
clc; clear; close all;

Nch = 32;
Rb = 200e9;             % bits/s/channel
M = 16;                 % 16-QAM
Np = 2;                 % dual polarization
alpha = 0.20;           % roll-off
grid = 50e9;            % Hz

% Symbol rate
Rs_16QAM = Rb/(Np*log2(M));
Rs_PAM4  = Rb/log2(4);

fprintf('DP-16QAM symbol rate = %.2f Gbaud\n', Rs_16QAM/1e9);
fprintf('PAM4 symbol rate = %.2f Gbaud\n', Rs_PAM4/1e9);

% Approximate occupied bandwidth
BW16 = (1+alpha)*Rs_16QAM;
BWP4 = (1+alpha)*Rs_PAM4;
fprintf('DP-16QAM bandwidth ~= %.2f GHz\n', BW16/1e9);
fprintf('PAM4 bandwidth ~= %.2f GHz\n', BWP4/1e9);

% Aggregate capacity
Ctotal = Nch*Rb;
fprintf('Aggregate capacity = %.2f Tb/s\n', Ctotal/1e12);

% Illustrative amplifier assumption:
% Pin/ch = -10 dBm, Pout/ch = 0 dBm
Pin_dBm = -10;
Pout_dBm = 0;
G = Pout_dBm-Pin_dBm;
Pout_W_ch = 10^((Pout_dBm-30)/10);
Ptotal_W = Nch*Pout_W_ch;
Ptotal_dBm = 10*log10(Ptotal_W/1e-3);

fprintf('Illustrative span gain = %.1f dB\n', G);
fprintf('Total output power = %.3f W = %.2f dBm\n', Ptotal_W, Ptotal_dBm);

% 50-GHz grid carrier positions
f0 = 193.1e12;
f = f0 + ((0:Nch-1)-(Nch-1)/2)*grid;

figure;
stem((1:Nch), f/1e12, 'filled');
```

```
grid on;
xlabel('Channel Number');
ylabel('Optical Frequency (THz)');
title('32-Channel DWDM Frequency Grid');

% Simplified Q-factor trend versus launch power
P_dBm = -8:0.25:4;
Q = 8 + 0.75*(P_dBm+8) - 0.55*(P_dBm+2).^2;
Q(Q<0)=0;

figure;
plot(P_dBm,Q,'LineWidth',1.5);
grid on;
xlabel('Launch Power per Channel (dBm)');
ylabel('Estimated Q-factor (a.u.)');
title('Illustrative Q-factor vs Launch Power');
```

Expected simulation interpretation: DP-16QAM should show a lower required baud rate than PAM4 for the same 200 Gb/s channel rate. The Q-factor versus launch power is expected to exhibit an optimum region: at low power, ASE/noise dominates; at excessive power, nonlinear penalties increase. The actual optimum and Q values must be obtained from the complete OptiSystem/OptSim link model.

FINAL ENGINEERING DECISION

YES – WITH TECHNICAL JUSTIFICATION.

The investigated DWDM architecture is recommended for engineering development because it reaches the required 6.4 Tb/s aggregate capacity using DP-16QAM at 25 Gbaud per channel, substantially reducing baud rate and spectral occupancy compared with PAM4. A high-speed EO MZI on silicon photonics provides a practical high-bandwidth integrated modulator, while the hybrid Raman–EDFA architecture addresses the long-haul requirement for optical gain and improved OSNR with gain flattening.

The coherent receiver performs CD equalization, adaptive polarization demultiplexing/PMD compensation, carrier recovery and nonlinear compensation. These functions directly address the major impairments listed in the assignment: CD, PMD, SPM, XPM, FWM, WDM crosstalk and ASE. The architecture also supports scalability through photonic integration and programmable DSP.

Deployment should be conditional on physical simulation and experimental validation of the actual span loss, amplifier noise figure, gain flatness, laser linewidth, modulator $V\pi$, fiber dispersion/nonlinearity parameters, receiver sensitivity, pre-FEC BER/Q-factor and thermal/power budget. The source assignment does not provide all numerical amplifier and fiber parameters, so the illustrative power calculation must be replaced with measured/vendor values before final deployment.

CONCLUSION

A 32-channel, 200 Gb/s/channel DWDM system requires careful coordination of modulation, photonic integration, amplification, fiber impairment management and coherent DSP. DP-16QAM provides the strongest overall compromise among spectral efficiency, baud rate and implementation complexity for the specified 200 Gb/s target. Silicon-photonic EO MZI modulation, AWG/MMI multiplexing, hybrid Raman–EDFA amplification, polarization management and coherent DSP form a technically consistent end-to-end architecture.

The analytical results establish 6.4 Tb/s aggregate capacity, 25 Gbaud DP-16QAM operation and an illustrative 15.05 dBm total amplifier output for 0 dBm/channel. Full OptiSystem/OptSim validation is required to replace the illustrative values with actual OSNR, BER, Q-factor, constellation and spectrum results.

SUBMISSION CHECKLIST

- Literature and standards review included in the technology-selection discussion.
- Modulation-format and baud-rate calculations included.
- Photonic modulator mechanism and architecture comparison included.

- Amplifier comparison and illustrative span-gain calculation included.
- Gain flattening and ASE accumulation discussed.
- CD, PMD, SPM, XPM, FWM, PDL and WDM crosstalk analyzed.
- Complete coherent DSP chain proposed.
- Integrated DWDM block diagram and design table included.
- MATLAB analytical model supplied.
- Numerical assumptions are explicitly identified where the source statement does not provide values.
- Final deployment decision and conditions for validation stated.